A rational counterexample to both Schur stability assertions

An exact specialization and Lean certificate for MathDB 370245

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Abstract

We verify a two-dimensional rational counterexample to Conjecture 1.1 of Banerjee and Chaudhury, arXiv:2509.06715v1. A primitive row-stochastic matrix W and a positive-definite matrix B meet every stated hypothesis, but both $W(I-tB)$ and $I-W+(I+tB)^{-1}(2W-I)$ have spectral radius exactly one at the strictly admissible step $t=1$. The example is the $s=1$ specialization of Shivam Patel's previously posted family. No mathematical novelty or priority is claimed. A single Lean module verifies the hypotheses, actual inverse, complete complex spectra, and failure of both stability assertions using mathlib's spectral radius.

1 The conjecture and its precise scope

For a real square matrix A, let $\sigma(A)$ be the spectrum of its complexification and define

$$\varrho(A) = \max\{|\lambda| : \lambda \in \sigma(A)\}.$$

The matrix is Schur stable if $\varrho(A) < 1$. A matrix W is row-stochastic if its entries are nonnegative and $We = e$, where e is the all-ones column vector. It is primitive if W^k has strictly positive entries for some positive integer k. We use the normalized left Perron vector π: $\pi > 0$, $\pi^\top e = 1$, and $\pi^\top W = \varrho(W)\pi^\top$.

Conjecture 1.1 of [1] asserts that, if W is primitive and stochastic and B is positive semidefinite with

$$Be \le \varrho(B)e, \qquad \pi^\top Be > 0, \tag{1}$$

then both operators

$$P(t) = W(I-tB), \qquad R(t) = I - W + (I+tB)^{-1}(2W-I) \tag{2}$$

are Schur stable for every $0 < t < 2/\varrho(B)$. All vector inequalities here are componentwise.

This conjecture does not impose entrywise nonnegativity on B. The paragraph immediately preceding it expressly discusses dropping that condition; Example 1.2 gives a matrix with negative off-diagonal entries and says that it satisfies the conjecture's assumptions. This distinction matters because the abstract and some earlier results of [1] discuss nonnegative matrices. Our target is the displayed Conjecture 1.1 with its own hypotheses.

2 The counterexample

Theorem 1. *Set*

$$W = \begin{pmatrix} 0 & 1 \\ 1/4 & 3/4 \end{pmatrix}, \qquad B = \frac{1}{11}\begin{pmatrix} 9 & -4 \\ -4 & 3 \end{pmatrix}, \qquad t = 1, \qquad \pi = \begin{pmatrix} 1/5 \\ 4/5 \end{pmatrix}. \tag{3}$$

Then W is primitive and row-stochastic, π is its normalized positive left Perron vector, and B is positive definite. Moreover,

$$\varrho(B) = 1, \qquad Be \le \varrho(B)e, \qquad \pi^\top Be = \frac{1}{55} > 0, \qquad 0 < t < \frac{2}{\varrho(B)}.$$

Nevertheless,

$$\sigma(P(1)) = \{0, 1\}, \qquad \sigma(R(1)) = \{1/2, 1\}.$$

Thus $\varrho(P(1)) = \varrho(R(1)) = 1$, and neither operator is Schur stable. Both assertions of Conjecture 1.1 are false.

Proof. The entries of W are nonnegative and its row sums are one. Also,

$$W^2 = \begin{pmatrix} 1/4 & 3/4 \\ 3/16 & 13/16 \end{pmatrix} > 0, \qquad \det(zI - W) = (z - 1)(z + 1/4).$$

Consequently W is primitive and $\varrho(W) = 1$. Direct multiplication gives $\pi^\top W = \pi^\top$. Both coordinates of π are positive and sum to one, so this is the required normalized left Perron vector.

The matrix B is symmetric. For $x = (x_1, x_2)^\top \in \mathbb{R}^2$,

$$11x^\top Bx = 8(x_1 - x_2/2)^2 + x_1^2 + x_2^2. \tag{4}$$

For $x \neq 0$ the last two terms have strictly positive sum, proving positive definiteness and hence positive semidefiniteness. Its characteristic polynomial factors as

$$\det(zI - B) = (z - 1)(z - 1/11).$$

Equivalently, $(2, -1)^\top$ and $(1, 2)^\top$ are eigenvectors with eigenvalues 1 and $1/11$, respectively. It follows that $\varrho(B) = 1$. Furthermore,

$$Be = \frac{1}{11}\begin{pmatrix} 5 \\ -1 \end{pmatrix} \le e, \qquad \pi^\top Be = \frac{5 - 4}{55} = \frac{1}{55} > 0.$$

Thus every condition in (1) holds, and $0 < 1 < 2/\varrho(B) = 2$.

For the first operator, exact multiplication gives

$$I - B = \frac{1}{11}\begin{pmatrix} 2 & 4 \\ 4 & 8 \end{pmatrix}, \qquad P(1) = \begin{pmatrix} 4/11 & 8/11 \\ 7/22 & 7/11 \end{pmatrix}. \tag{5}$$

Its trace is one and its determinant is zero, so $\det(zI - P(1)) = z(z - 1)$. More concretely,

$$P(1)\begin{pmatrix} 8 \\ 7 \end{pmatrix} = \begin{pmatrix} 8 \\ 7 \end{pmatrix}.$$

This nonzero eigenvector already obstructs strict Schur stability.

For the second operator, put

$$J = \begin{pmatrix} 7/12 & 1/6 \\ 1/6 & 5/6 \end{pmatrix}.$$

Direct multiplication verifies both $J(I+B) = I$ and $(I+B)J = I$; therefore $J = (I+B)^{-1}$, the actual inverse appearing in (2). For additional transparency,

$$I - W = \begin{pmatrix} 1 & -1 \\ -1/4 & 1/4 \end{pmatrix}, \qquad J(2W - I) = \begin{pmatrix} -1/2 & 5/4 \\ 1/4 & 3/4 \end{pmatrix}.$$

Adding them yields

$$R(1) = \begin{pmatrix} 1/2 & 1/4 \\ 0 & 1 \end{pmatrix}. \tag{6}$$

This triangular matrix has characteristic polynomial $(z-1/2)(z-1)$; indeed $R(1)(1,2)^\top = (1,2)^\top$. The displayed factorizations hold for every complex z and identify the complete complex spectra, not just real roots. The two spectral radii are therefore exactly one. This proves all assertions. □

The step $t = 1$ lies strictly inside the conjectured open interval $(0, 2)$. The counterexample uses an eigenvalue on the unit circle, which suffices because the conjecture requires a strict inequality. No eigenvalue strictly larger than one, numerical root approximation, or limiting argument is needed.

3 Attribution and limits of the contribution

Patel's reply [2] already presents the family

$$W_s = \begin{pmatrix} 0 & 1 \\ v & 1-v \end{pmatrix}, \qquad v = \frac{1}{5-s}, \qquad 0 < s < 2,$$

with the same B as in (3). The posted calculation states that both operators have eigenvalue one at $t = s$ and a real eigenvalue exceeding one for $s < t < 2$. Substituting $s = 1$ gives exactly our W and t; normalizing $(v, 1)^\top$ gives exactly our π.

Our contribution is an independent exact verification and a Lean formalization of this specialization, supplied for the MathDB 370245 verification campaign. We do not claim discovery of the counterexample, a new mathematical theorem, the first formalization, or a verification of every assertion about the full posted family. The broader family is not needed here. We do not challenge the separate local or special-case results in [1]; failure at one admissible step does not contradict a per-instance sufficiently-small-step theorem.

4 Formal certificate and reproduction

The accompanying module `formal/SchurStability.lean` uses Lean 4.34.1 and pinned mathlib revision

`d13f23b723b8a846827a245b89c10fc7d3f11612`

It defines W, B, t, π as the literal real data in (3), and defines P and R by (2). In particular, R uses matrix inversion; it is not defined by the simplified answer (6).

For a real matrix A, `complexify A` maps its entries through $\mathbb{R} \hookrightarrow \mathbb{C}$. The definition `rho A` is mathlib's `spectralRadius` over $\mathbb{C}$ of this complex matrix. Its codomain is the extended

nonnegative reals. The definition `realRho A` takes its real value. The four exact radius-one proofs in this example make that conversion harmless; no infinite value is used as a real radius.

The spectrum calculation is connected to mathlib by the equivalence

$$z \in \sigma(A) \quad \Longleftrightarrow \quad \det(zI - A) = 0$$

for complex matrices. The library definition characterizes the left side by noninvertibility; invertibility of a matrix is equivalent to invertibility of its determinant. Over $\mathbb{C}$, the latter fails exactly when the determinant is zero. The certificate proves this bridge, computes the four factorizations, and uses

$$\mathrm{spectralRadius}_{\mathbb{C}}(A) = \sup_{z \in \sigma(A)} \|z\|$$

to obtain the exact radii. It also formally identifies matrix spectra with finite-dimensional linear-map eigenvalues, and verifies nonzero complex eigenvectors of $P(1)$ and $R(1)$ with eigenvalue one.

Mathematical content	Lean declarations (namespace `SchurStability`)
Stochasticity and primitivity	`W_stochastic`, `W_square`, `W_primitive`
Perron vector	`pi_positive`, `pi_normalized`, `pi_is_left_Perron`
Positive definiteness	`B_quadratic`, `B_posDef`, `B_posSemidef`
Side conditions and strict step	`B_source_bound`, `pi_B_e_positive`, `strict_step`
Actual inverse and operators	`inverse_left`, `inverse_right`, `actual_inverse`, `P_exact`, `R_exact`
Complete complex spectra	`B_spectrum`, `W_spectrum`, `P_spectrum`, `R_spectrum`
Exact spectral radii	`rho_B`, `rho_W`, `rho_P`, `rho_R`
Eigenvector/eigenvalue bridge	`eigenvalue_iff_spectrum`, `P_complex_eigenvector`, `R_complex_eigenvector`
All hypotheses and both failures	`schur_counterexample`
Negated universal assertions	`both_conjectures_false`

The last declaration refutes both universal statements already in dimension two, which is enough to refute their unrestricted versions. The source hypotheses include normalization of the Perron vector, a harmless strengthening of the source's choice of scaling.

From an extracted release directory, run

```
python3 verify_outcome.py .
```

The campaign verifier checks pinned dependencies, compiles a byte-identical isolated source copy with warnings treated as errors, and prints the type and foundational dependencies of every declaration. The release includes the verifier and dependency pins. The completed audit checks 77 declarations and reports only `propext`, `Classical.choice`, and `Quot.sound`. An independent review accompanies the release; it is separate from kernel checking.

References

[1] D. Banerjee and K. N. Chaudhury, *On the Schur Stability of Some Image Reconstruction Operators*, arXiv:2509.06715v1, 8 September 2025. https://arxiv.org/html/2509.06715v1. Conjecture 1.1, its preceding paragraph, and Example 1.2.

[2] S. Patel (username `shivampatel`), *Counterexample to both stability assertions, with arbitrarily small instability thresholds*, MathDB 370245, reply 1, 20 August 2026. Frozen reply identifier `5a30dc24-61b4-49cc-a6de-ee5f67a71bf3`, recorded in `metadata.json`.